VoiceOps 2026: Grounded Customer Support Voice AI for ZentroMart

Capstone Project Report · LLM & Generative AI Course
Consulting & Analytics Club, IIT Guwahati × Gnani.ai
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Source code (GitHub)	https://github.com/Shikha07-Git/VoiceOps-2026
Demo video (max 2 min)	Google Drive link
Dataset	Google Drive link

1. Project Overview

VoiceOps 2026 is an AI customer support system for ZentroMart, a simulated Indian e-commerce retailer. It combines a large language model, retrieval-augmented generation (RAG), conversation state, mock business tools and speech processing. The assistant helps with order tracking, cancellations, returns, refunds, warranty queries and escalation requests, and grounds its answers in the company documentation.

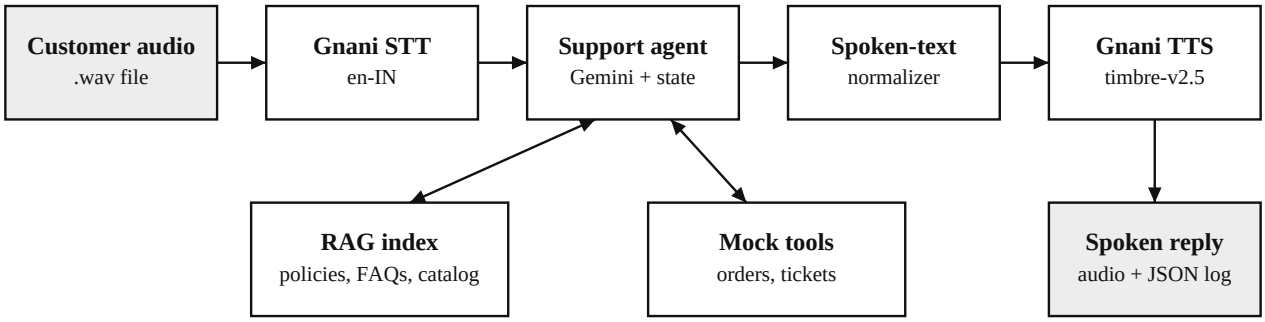


Figure 1. System architecture.

Key features

- Retrieval from policies, FAQs, help articles and the product catalog, with active policy ranked above superseded documents.
- Multi-turn conversations with identity verification (phone-number suffix or delivery PIN code) before protected order information is disclosed.
- Mock order-lookup and support-ticket tools, and deterministic business-day refund calculations that account for holidays.
- Gnani.ai speech-to-text and text-to-speech, a browser dashboard, and a batch audio runner.

2. Development Milestones

Step 1: prompted LLM baseline

A Gemini language model with a customer-support persona and no retrieval from company documents. On 52 practice questions it answered 17 correctly (32.69%). General model knowledge produced inaccurate return periods and refund timelines for company-specific policies.

Step 2: knowledge grounding with RAG

The Markdown policy documents and their metadata are parsed and chunked by section, and the product catalog is indexed alongside them. Chunks are embedded with `gemini-embedding-001` and retrieved by cosine similarity weighted by document priority, with extra matching for product identifiers and policy categories. The answer is generated from the retrieved context. The system answered 42 of 52 questions correctly (80.77%).

Step 3: multi-turn conversations and tool calling

A stateful agent recognizes order numbers, verifies the caller, then gives verified access to order details and keeps context across turns. It calls the mock lookup and ticket tools, creates support tickets for escalations and answers follow-ups from policy. It passed 86 of 96 evaluated turns (89.58%) across 23 conversations.

Step 4: voice integration

Gnani.ai speech-to-text converts customer audio to text and Gnani.ai text-to-speech speaks the reply. A normalization step formats dates, amounts and order identifiers for clear speech. The batch runner `run.py` processes audio files and saves reply audio and structured JSON logs.

3. Evaluation Results

Metric	Result
Step 1: LLM baseline	32.69% (17 / 52)
Step 2: grounded RAG	80.77% (42 / 52)
Improvement	+48.08 percentage points
Step 2 retrieval hit rate*	96.15% (50 / 52)
Step 3 multi-turn evaluation	89.58% (86 / 96 turns), 23 conversations

Table 1. Results on the practice questions and conversations.

*The retrieval hit rate follows the evaluation script's definition and counts questions that have no specified reference evidence, so it should not be read as retrieval precision. All scores come from automated matching criteria on practice scenarios; they do not establish human-rated answer quality or freedom from hallucination. The result files are in `sample_outputs/`.

4. Known Limitations and Future Improvements

Observed issue	Planned improvement
Refund answers sometimes omit exact dates	Integrate the deterministic timeline results directly into the final response
Some escalation scenarios are missed	Enforce escalation rules before the model generates a reply
A product warranty duration was answered incorrectly	Give verified SKU-level metadata stronger priority
Some non-returnable product queries fail	Improve category normalization and policy matching

Table 2. Issues seen in evaluation and the planned changes.

Further work: human evaluation of answers, more regression tests, better confidence calibration, and wider voice testing.

5. Bonus Work

- Interactive web dashboard (`web_app.py`): typed or spoken input, audio playback, and a panel showing verification status, order context, retrieved policy references and ticket details.
- Batch voice runner (`run.py`): reads audio from an input folder and writes reply audio and JSON results to an output folder.
- Scripts for voice testing, sample-audio preparation and local diagnostics.

6. Technology Stack

Component	Technology
Language, model	Python; Google Gemini
Embeddings and retrieval	`gemini-embedding-001`; dense similarity with policy-aware scoring
Speech	Gnani.ai speech-to-text and text-to-speech
Dashboard	Python HTTP server, HTML, CSS, JavaScript
Business tools and evaluation	Python mock order and ticket tools; custom evaluation scripts

Table 3. Technology stack.

7. Setup and Usage

Clone the repository, install the dependencies and create the environment file:

```
git clone https://github.com/Shikha07-Git/VoiceOps-2026.git
cd VoiceOps-2026
python -m pip install -r requirements.txt
Copy-Item .env.example .env
```

Add the Gemini and Gnani.ai keys to `.env` (it is excluded from Git). Extract the capstone dataset into the project root as `Capstone_Project_Dataset_FTTT_2026/` with its `data`, `kb`, `practice` and `tools` folders; it is not part of the public repository. Then run:

```
python web_app.py          # dashboard at http://localhost:8080
python run.py --input_dir sample_audio --output_dir batch_test_outputs
python -m src.step1_baseline
python -m src.step2_rag_eval
python -m src.step3_conv_eval
```

Real speech processing needs valid Gnani.ai credentials and compatible audio. Re-running the evaluations can give different results because of model behavior and API changes. This is a capstone prototype, not a production service; deployment would need stronger authentication, privacy controls, monitoring and security testing.

8. Conclusion

Grounding the assistant in company documents raised practice-question accuracy from 32.69% to 80.77% under the project's automated criteria. The remaining challenges are exact refund dates, consistent escalation and product-specific policy interpretation.